UNIT 3.
Lectures 3.2, 3,3 Molecular composition of membranes



Pérez-Medavilla → → ¶

The·composition·and·architecture·of·membranes¶

¶

Structural·lipids·in·membranes·¶

·Plasma·membranes·act·as·intelligent·barriers·that·isolate·cells,·in·such·a·way· that·it·is·the·cell·itself·that·decides·which·extracellular·signals·can·reach·the· interior·of·it.¶

·Likewise,·the·organelle·membranes·allow·the·cell·to·be·compartmentalized,·in· such·a·way·that·substances·can·accumulate·inside·it,·which·are·released·in·a· controlled·way·to·the·cytoplasm,·or·separate·metabolic·processes.·¶

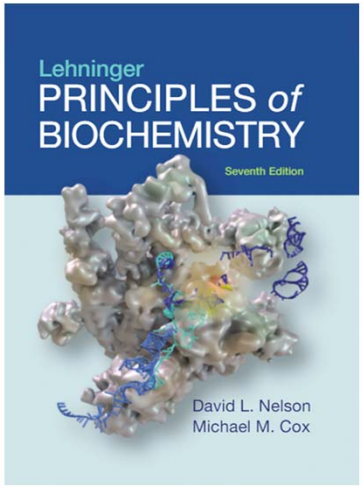
·The·structural·characteristic·of·membranes·is·that·they·are·formed·by·a·lipid· bilayer.·¶

·In·order·to·understand·the·function·of·membranes,·it·is·necessary·to·know·their· molecular·composition:·polar·lipids,·proteins·and·carbohydrates·bound·to· proteins·and·lipids.¶

·Common·to·all·membrane·lipids·is·that·they·are·amphipathic·molecules,·that· is,·one·end·is·hydrophilic·and·the·other·hydrophobic.·¶

·The·different·membrane·lipids·are:¶

- ·Lipids·with·phosphate:·Glycerophospholipids·and·Sphingomyelins¶
- ·Lipids·with·sugars·glycerophospholipids,·cerebrosides·and·gangliosides¶
- ·Cholesterol¶



Additional study material

ADI document:

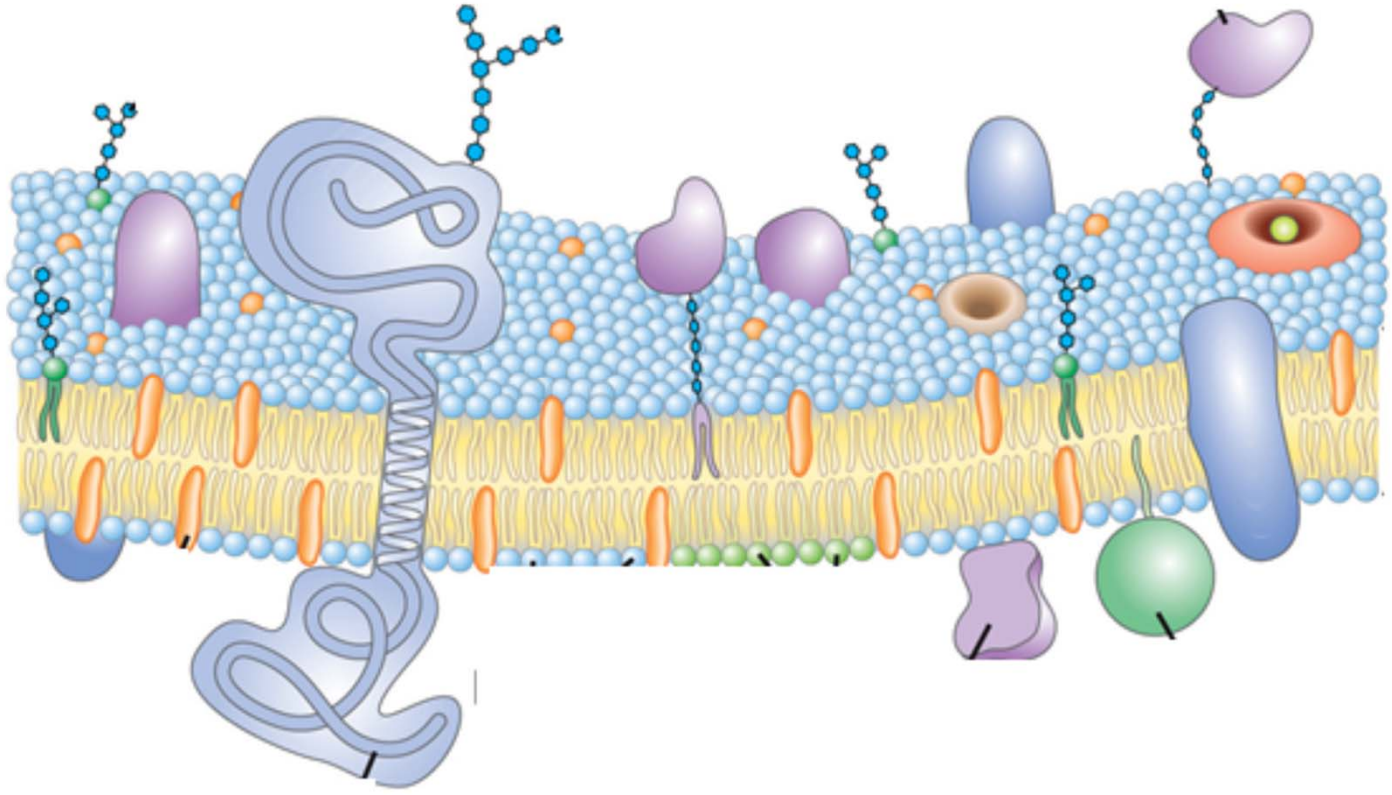
Molecular structure of membranes

Reading

Lehninger's PoB Chapter 11:

11.1 The composition and architecture of membranes

Molecular Composition of membranes



Functions of Membranes

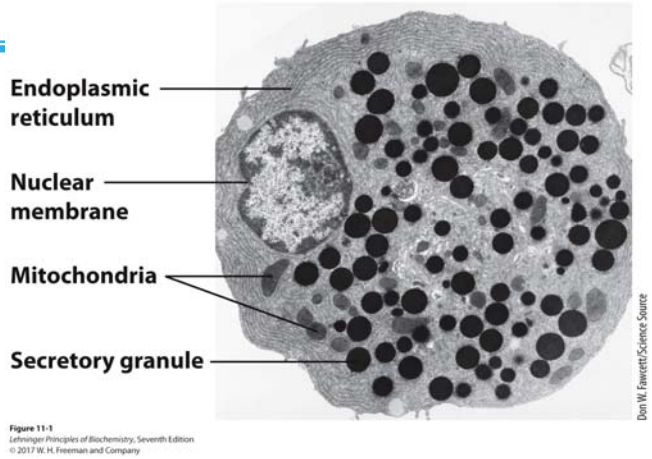
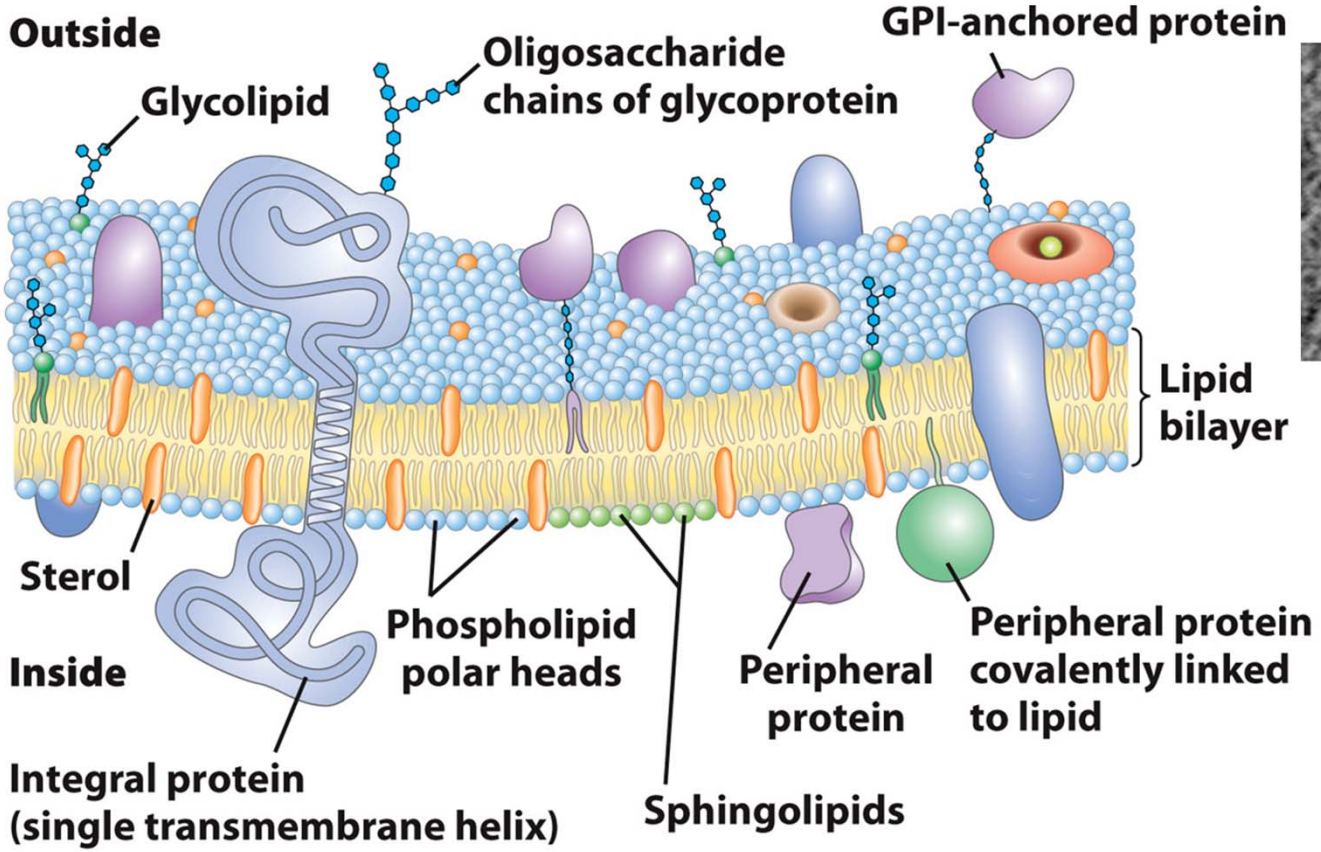


Figure 11-1
Lehninger Principles of Biochemistry, Seventh Edition
© 2017 W. H. Freeman and Company

- Define the boundaries of the cell
- Allow import and export
  - Selective import of nutrients (e.g. lactose)
  - Selective export of waste and toxins (e.g. antibiotics)
- Retain metabolites and ions within the cell
- Sense external signals and transmit information into the cell
- Provide compartmentalization within the cell
  - separate energy-producing reactions from energy-consuming ones
  - keep proteolytic enzymes away from important cellular proteins
- Produce and transmit nerve signals
- Store energy as a proton gradient
- Support synthesis of ATP

Fluid mosaic model for plasma membrane structure



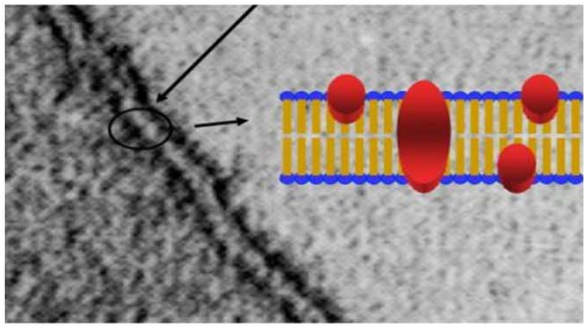


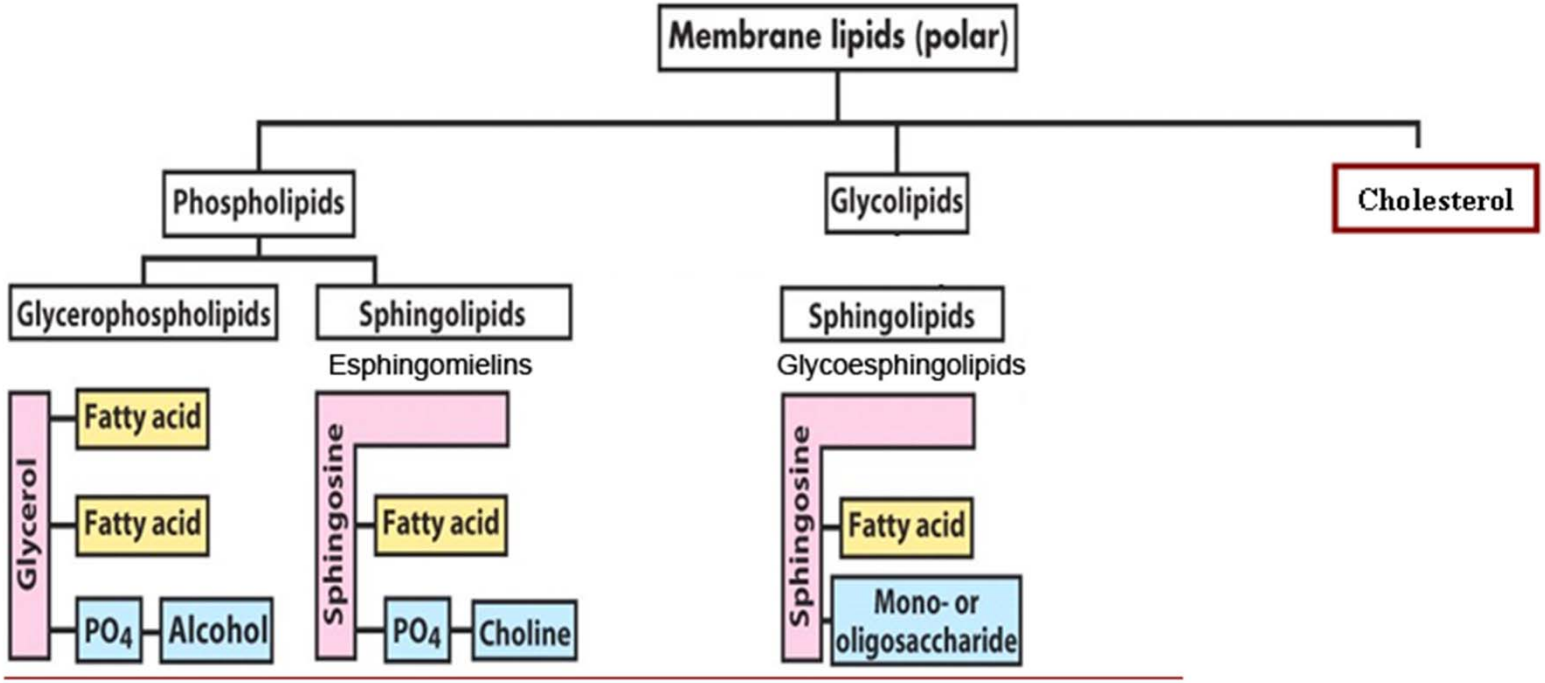
Figure 11-3
Lehninger Principles of Biochemistry, Seventh Edition

Common Features of Membranes

- Sheet-like flexible structure, 30–100 Å (3–10 nm) thick
- Main structure is composed of two leaflets of lipids (bilayer)
- Form spontaneously in aqueous solution and are stabilized by noncovalent forces, especially hydrophobic effect
- Protein molecules span the lipid bilayer
- Asymmetric
  - Some lipids are found more commonly “inside.”
  - Some lipids are found more commonly “outside.”
  - Carbohydrate moieties are attached on the outer leaflet.
  - They can be electrically polarized.
- Fluid structures: two-dimensional solution of oriented lipids

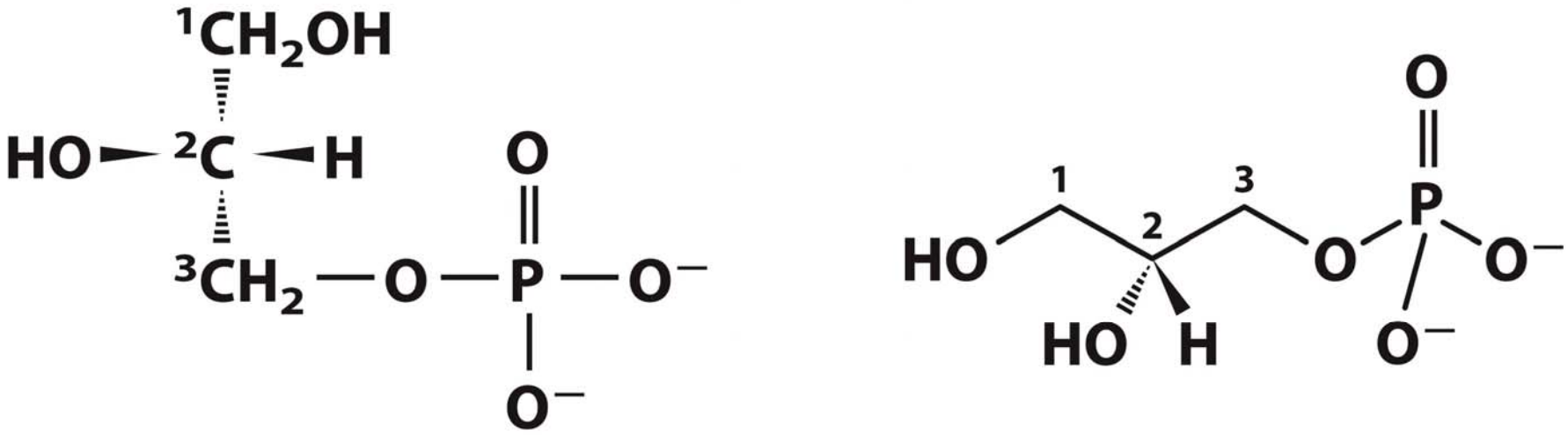
Types of Lipids in the membranes

- Contain polar head groups and nonpolar tails (usually attached fatty acids)
- Diversification can come from:
  - modifying a different backbone
  - changing the fatty acids
  - modifying the head groups



Glycerophospholipids

- Primary constituents of cell membranes
- Two fatty acids form ester linkages with the first and second hydroxyl groups of L-glycerol-3-phosphate
- Head group is charged at physiological pH



L-Glycerol 3-phosphate (*sn*-glycerol 3-phosphate)

Figure 10-8
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

General Structure of Glycerophospholipids

- Unsaturated fatty acids are commonly found connected to C2
- The highly polar phosphate group may be further esterified by an alcohol; such substituent groups are called the head groups

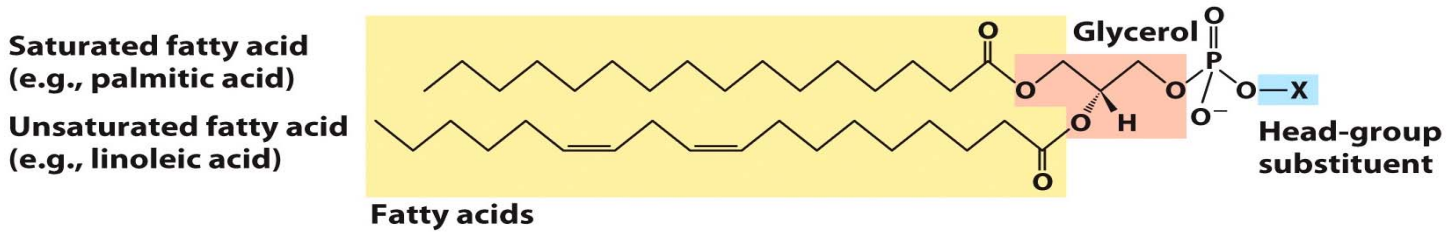


Figure 10-9
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

Examples of Glycerophospholipids

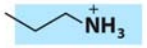
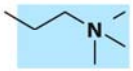
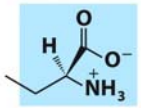
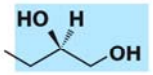
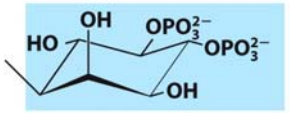
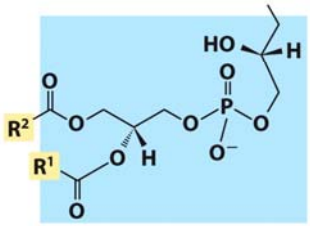
Name of glycerophospholipid	Name of X—O	Formula of X	Net charge (at pH 7)
Phosphatidic acid	—		−2
Phosphatidylethanolamine	Ethanolamine		0
Phosphatidylcholine	Choline		0
Phosphatidylserine	Serine		−1
Phosphatidylglycerol	Glycerol		−1
Phosphatidylinositol 4,5-bisphosphate	*myo*-Inositol 4,5-bisphosphate		−4*
Cardiolipin	Phosphatidyl-glycerol		−2

Figure 10-9
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

Phosphatidylcholine

- Phosphatidylcholine is the major component of most eukaryotic cell membranes

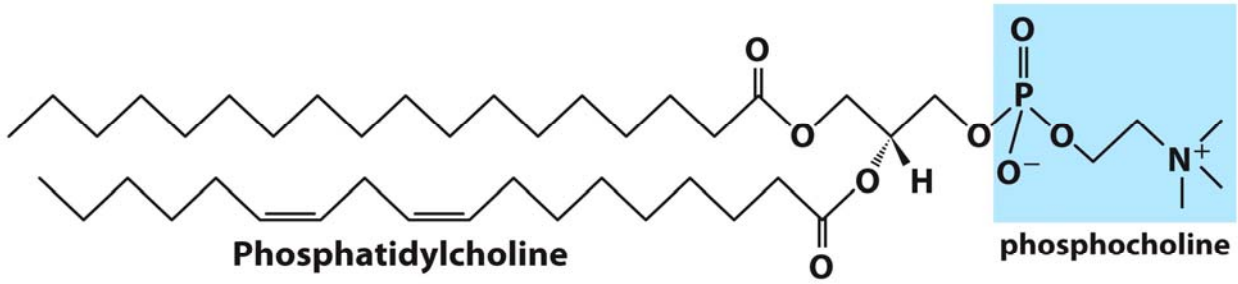
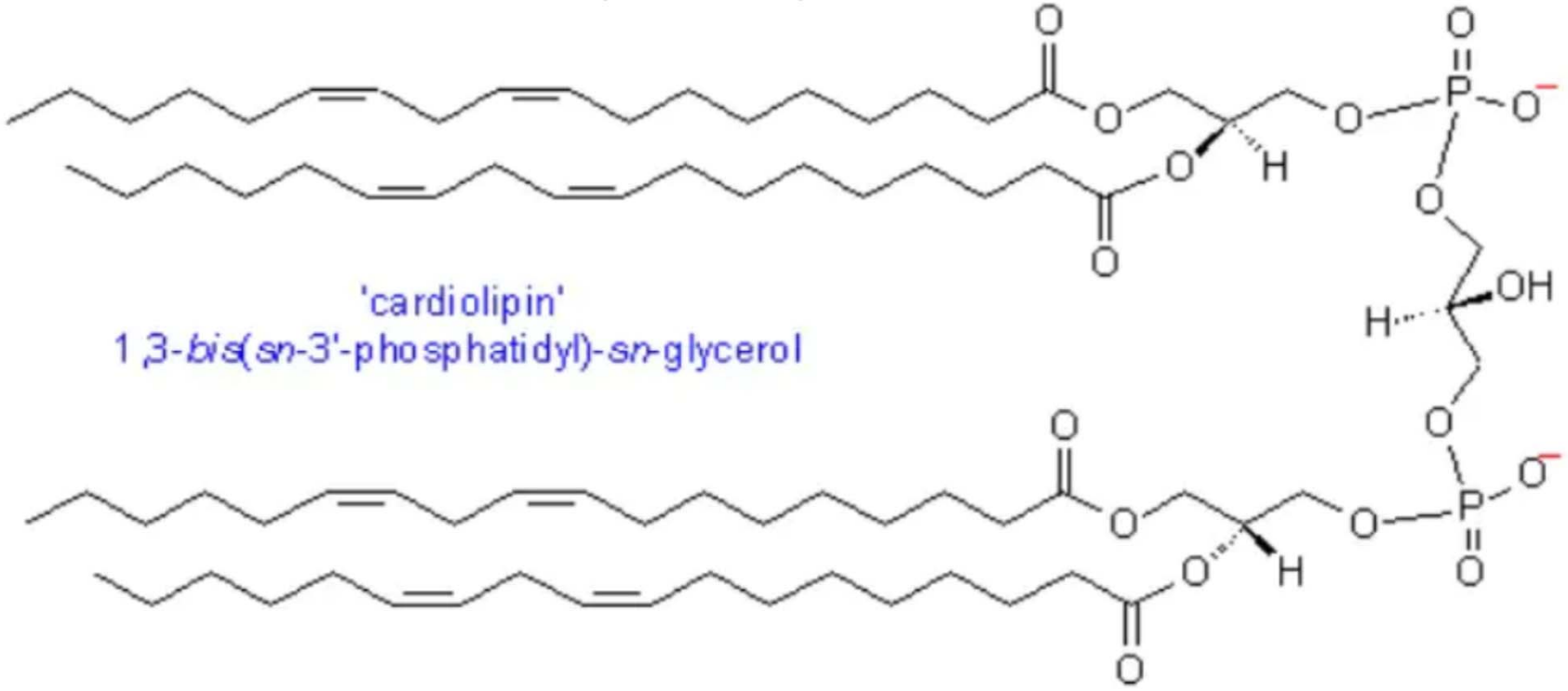


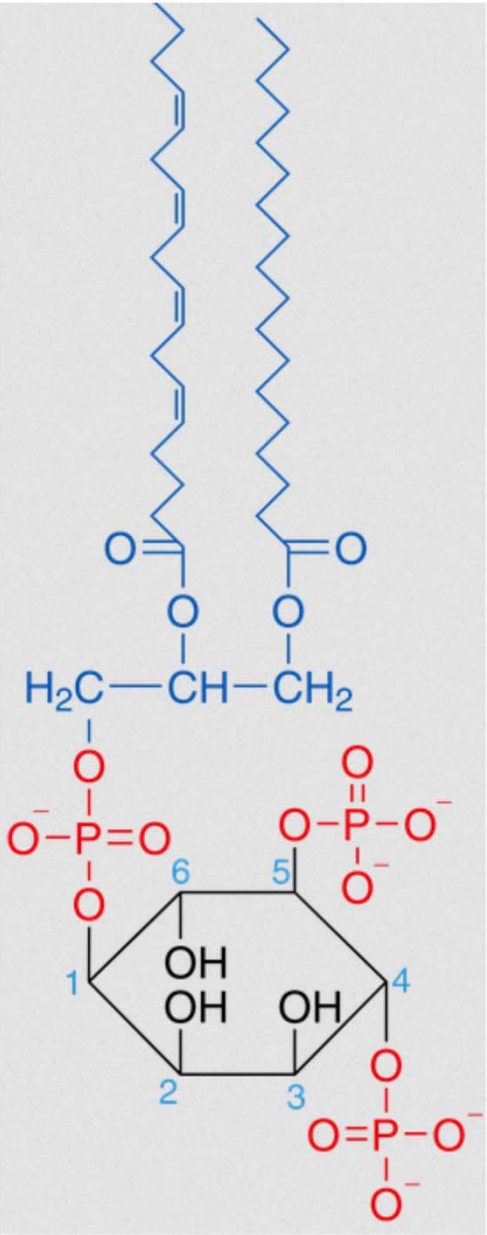
Figure 10-14
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

Cardiolipin

- Mainly located at the inner mitochondrial membrane
- Very abundant in the cardiac tissue
- Necessary for the activity of various enzymes, especially in the mitochondrial respiratory chain



Phosphatidylinositol



- Can be specifically phosphorylated at various OH
- It is a precursor of molecules with an important role in intracellular signal transduction (DAG and IP3)

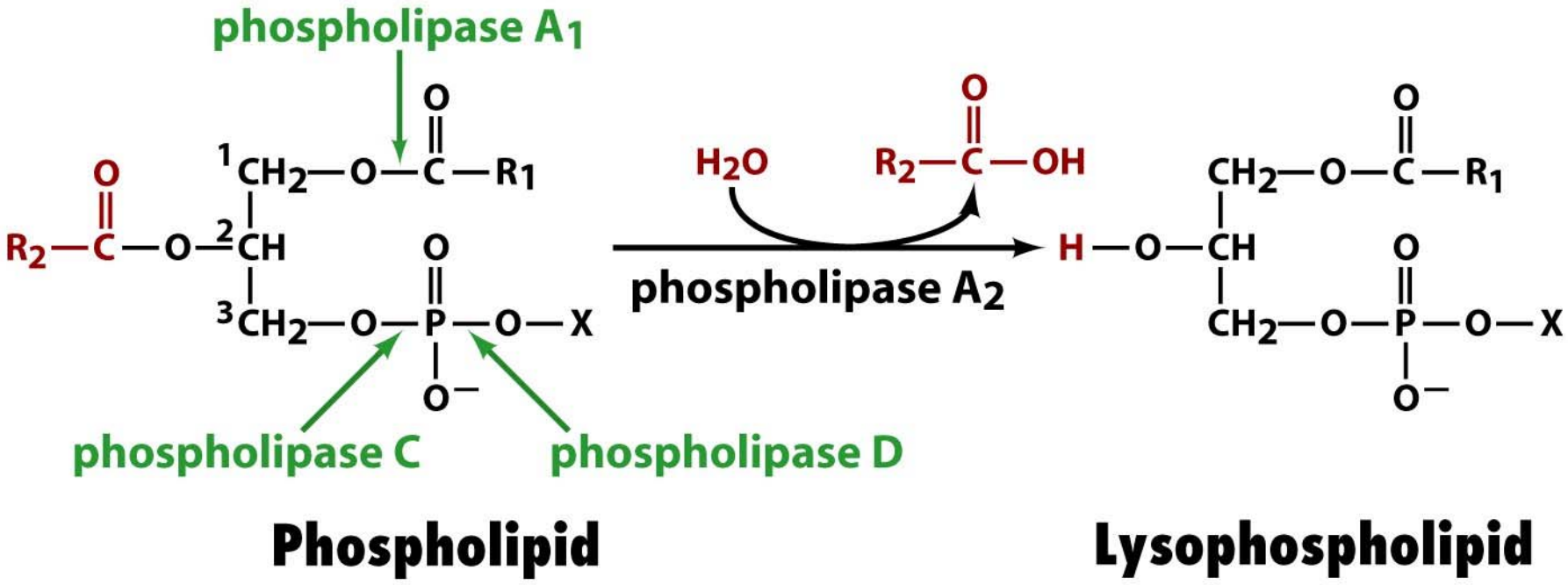
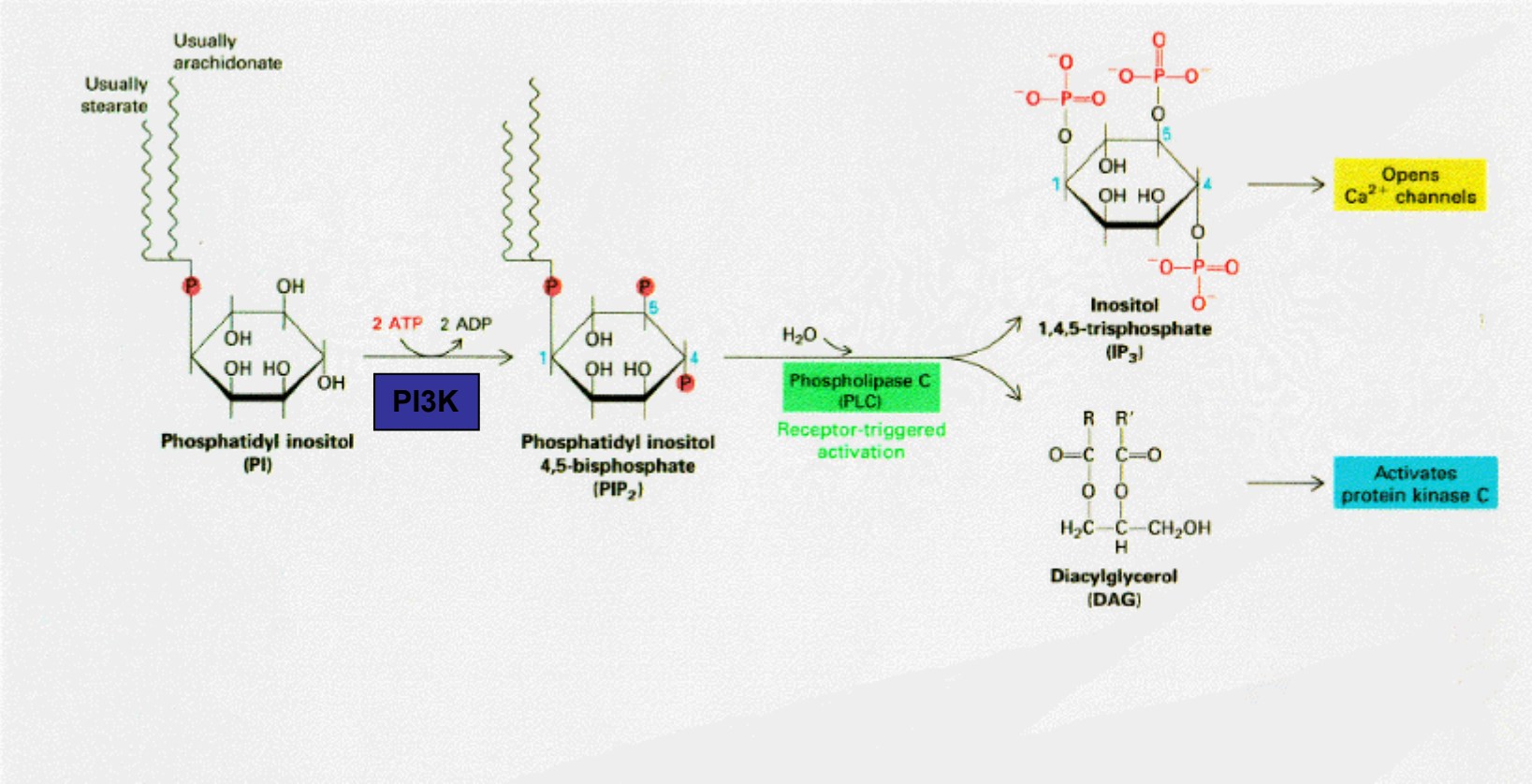


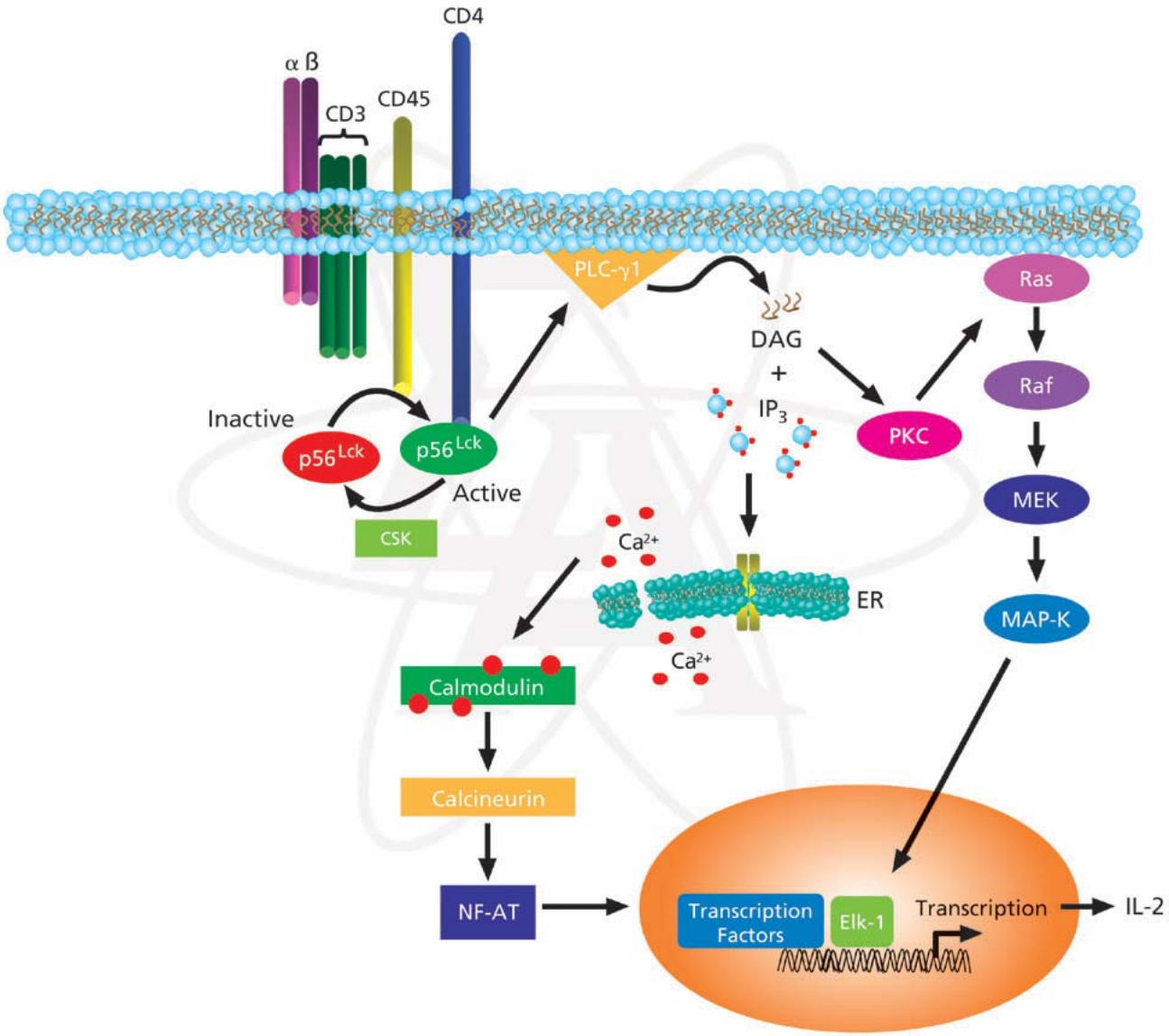
Figure 9-5 Fundamentals of Biochemistry, 2/e
© 2006 John Wiley & Sons

Membrane Phospholipids are source of important second messengers after enzyme-mediated hydrolisis

Phosphoinositides signaling



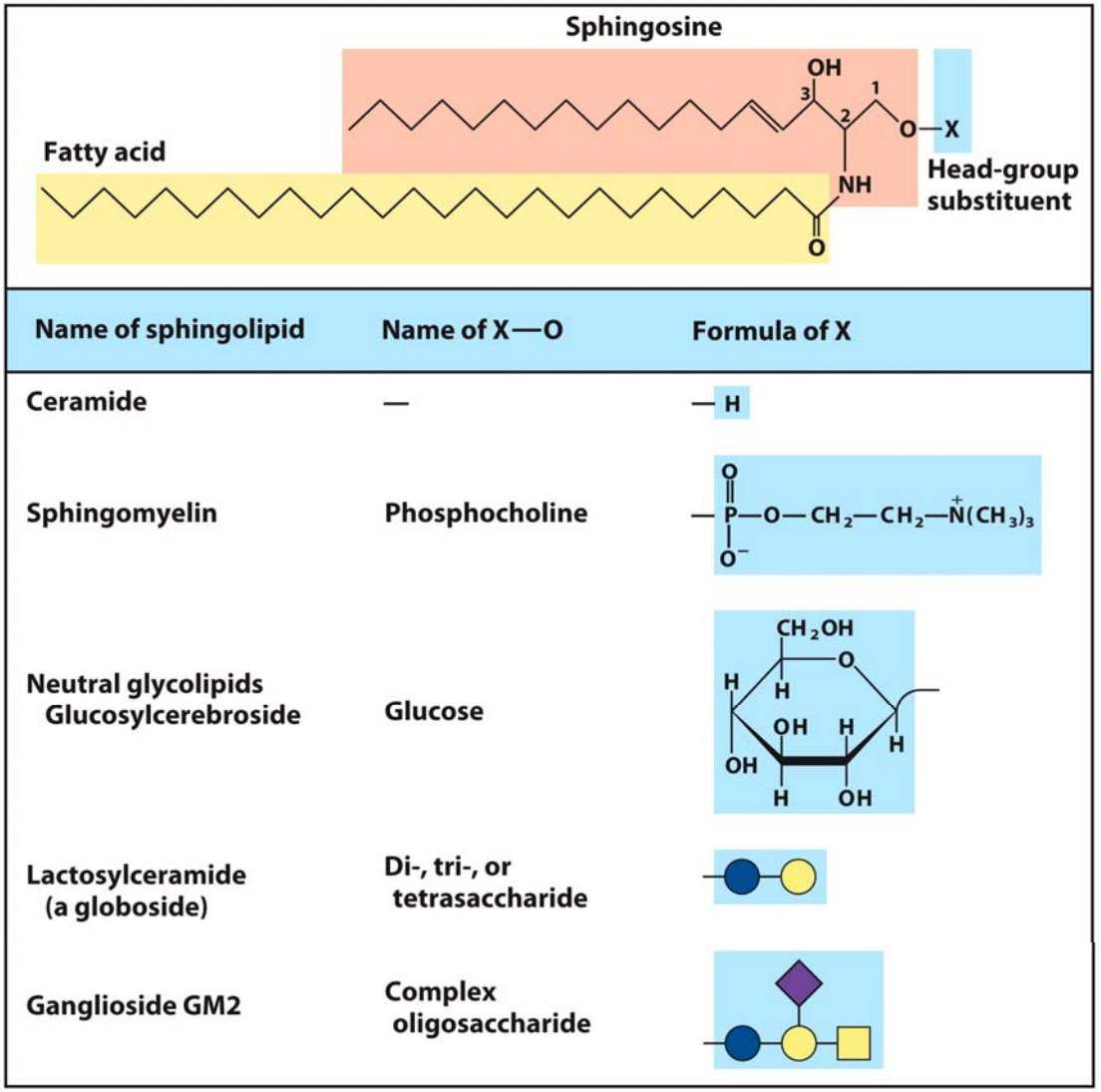
T Cell Receptor (TCR) Signaling



Sphingolipids

- The backbone of sphingolipids is NOT glycerol
- The backbone of sphingolipids is a long-chain amino alcohol sphingosine
- A fatty acid is joined to sphingosine via an amide linkage rather than an ester linkage as usually seen in lipids
- A polar head group is connected to sphingosine by a glycosidic or phosphodiester linkage
- The sugar-containing glycosphingolipids are found largely in the outer face of plasma membranes

Sphingolipids



Name of sphingolipid	Name of X—O	Formula of X
Ceramide	—	— H
Sphingomyelin	Phosphocholine	
Neutral glycolipids Glucosylcerebroside	Glucose	
Lactosylceramide (a globoside)	Di-, tri-, or tetrasaccharide	
Ganglioside GM2	Complex oligosaccharide	

Figure 10-13
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

Sphingomyelin

- Ceramide (sphingosine + amide-linked fatty acid) + phosphocholine attached to the alcohol
- Sphingomyelin is abundant in myelin sheath that surrounds some nerve cells in animals

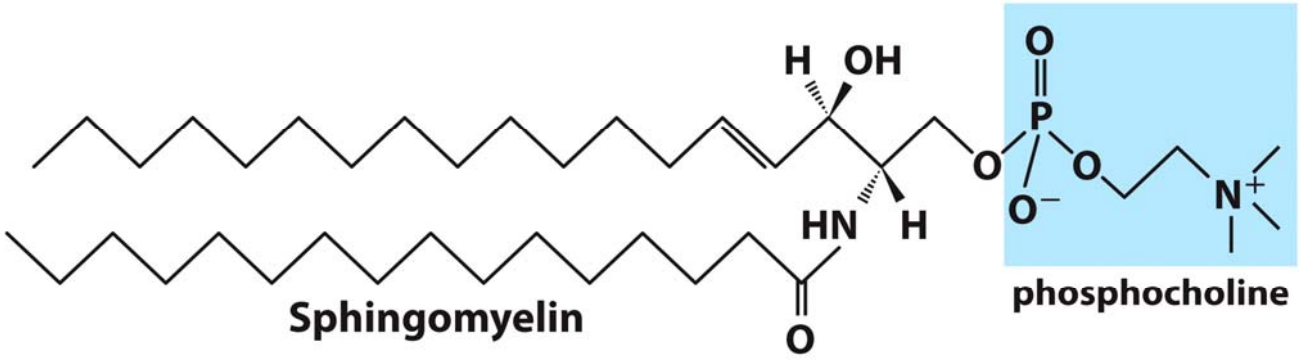


Figure 10-14
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

Sphingomyelin is structurally similar to phosphatidylcholine

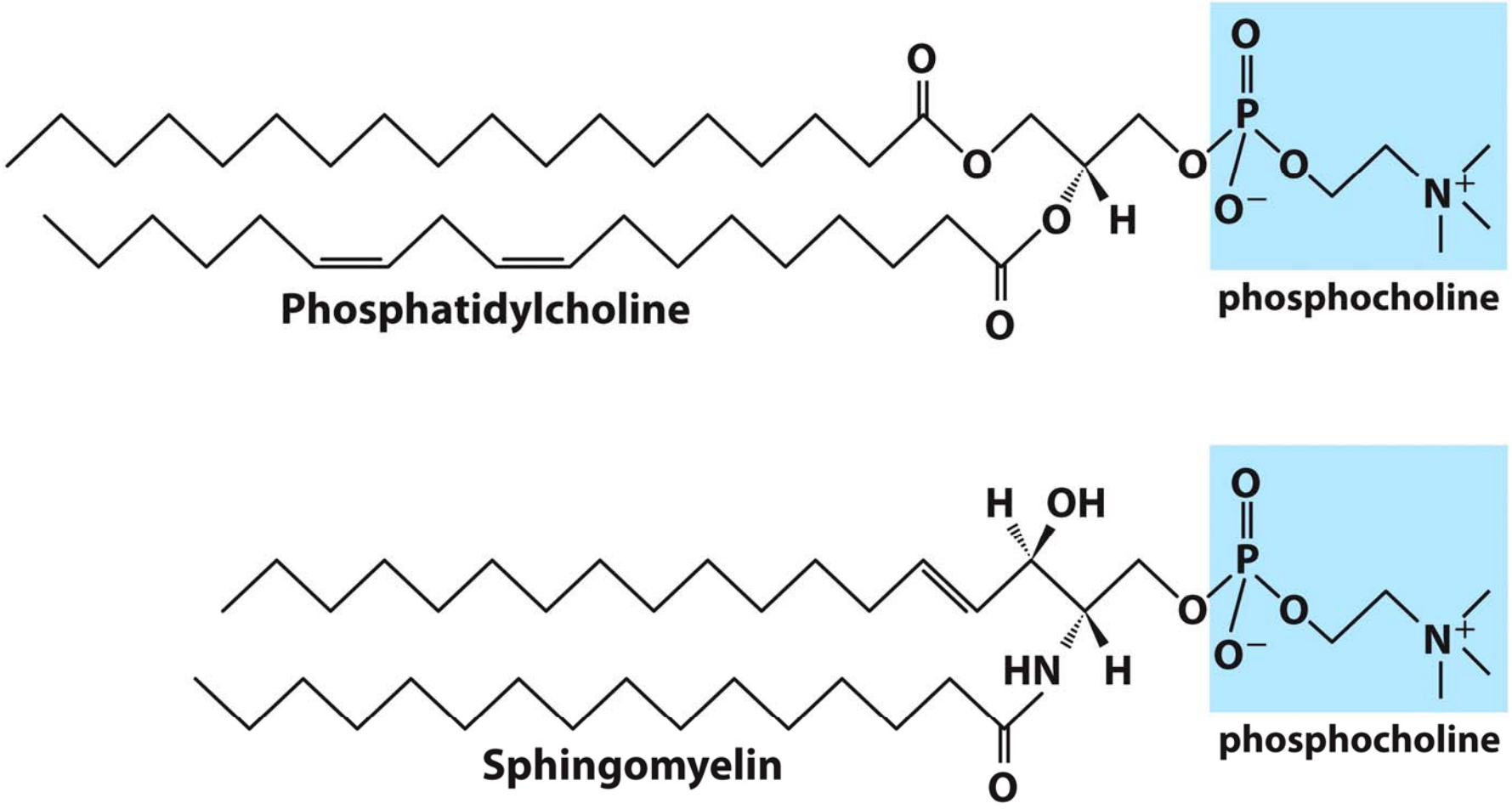
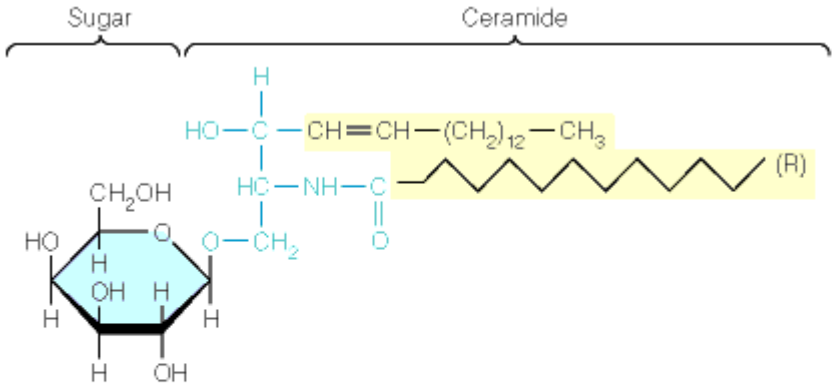


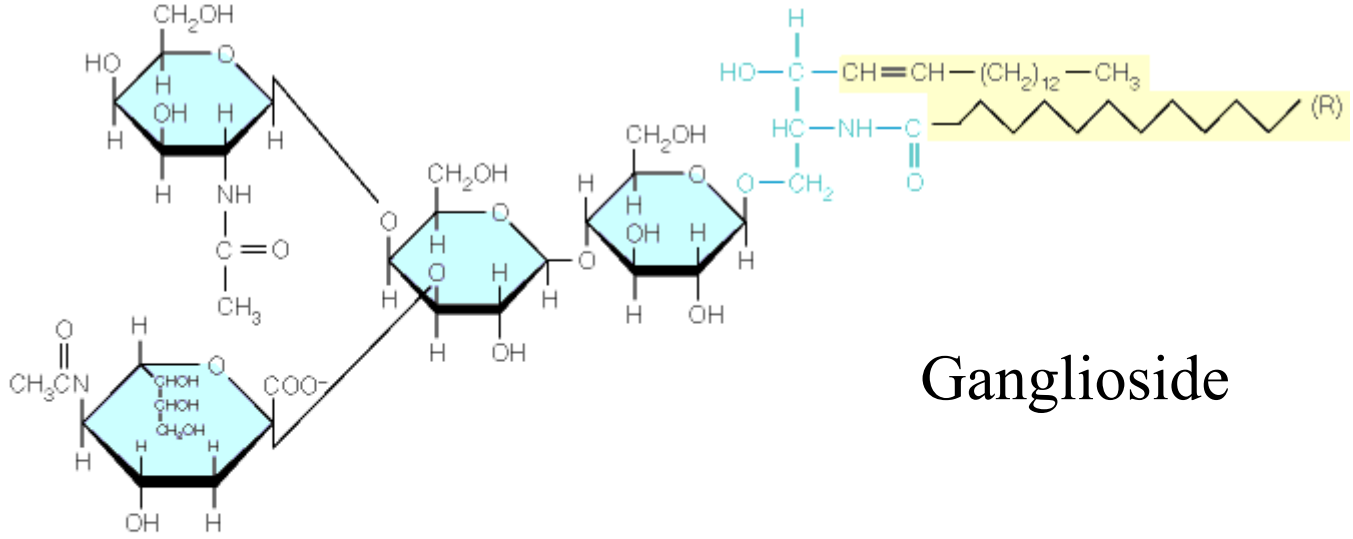
Figure 10-14
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

Glicosphingolipids



(a) Galactosylceramide

Cerebroside



(b) GalNAcβ(1→4)Galβ(1→4)Glcβ(1→1)ceramide

(3 ↑ α2 Sia)

Ganglioside

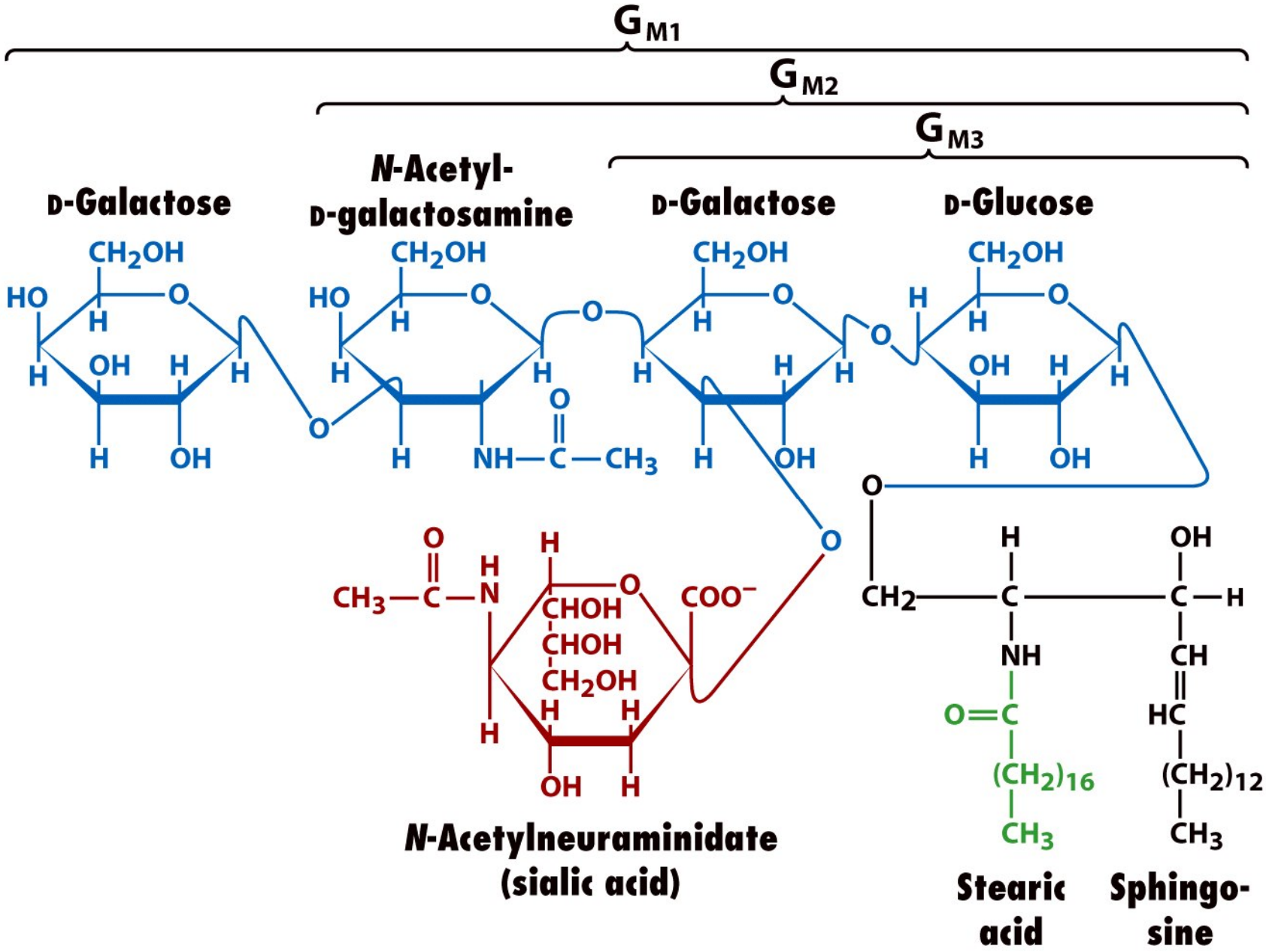


Figure 9-9a Fundamentals of Biochemistry, 2/e

© 2006 John Wiley & Sons

Glycosphingolipids and Blood Groups

- The blood groups are determined in part by the type of sugars located on the head groups in glycosphingolipids.
- The structure of sugar is determined by an expression of specific glycosyltransferases
  - Individuals with no active glycosyltransferase will have the **O antigen**
  - Individuals with a glycosyltransferase that transfers an N-acetylgalactosamine group have **A blood group**
  - Individuals with a glycosyltransferase that transfers a galactose group have **B blood group**

Glycosphingolipids determine blood groups

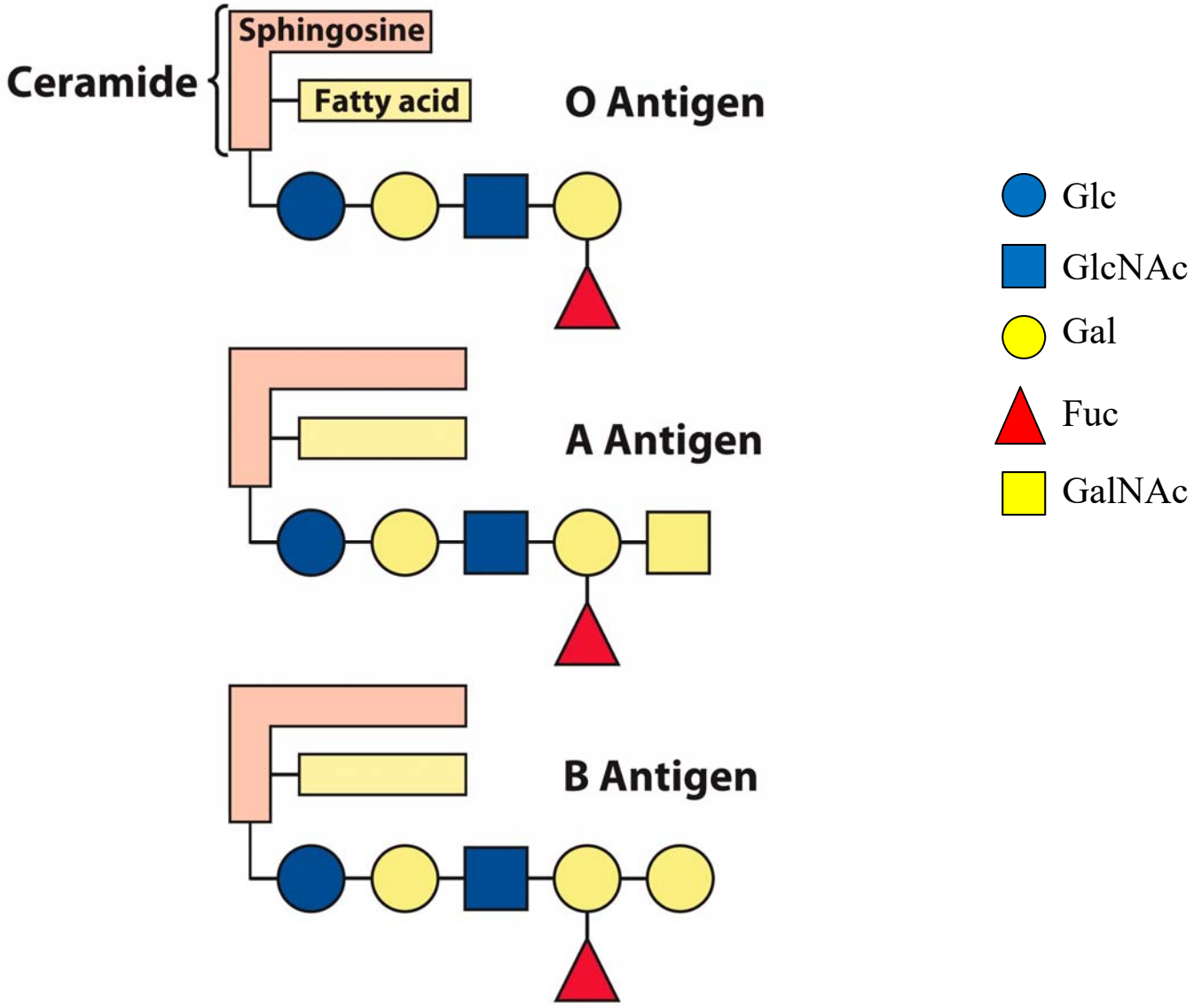
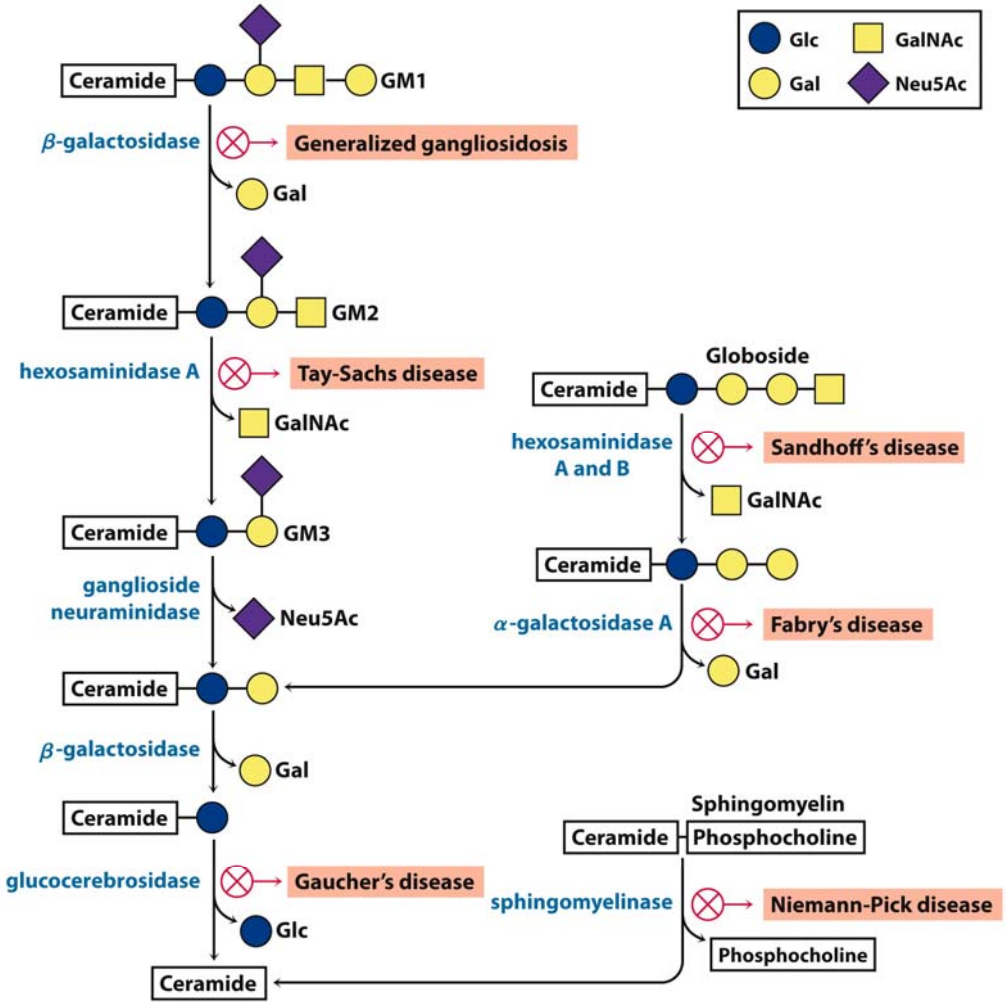


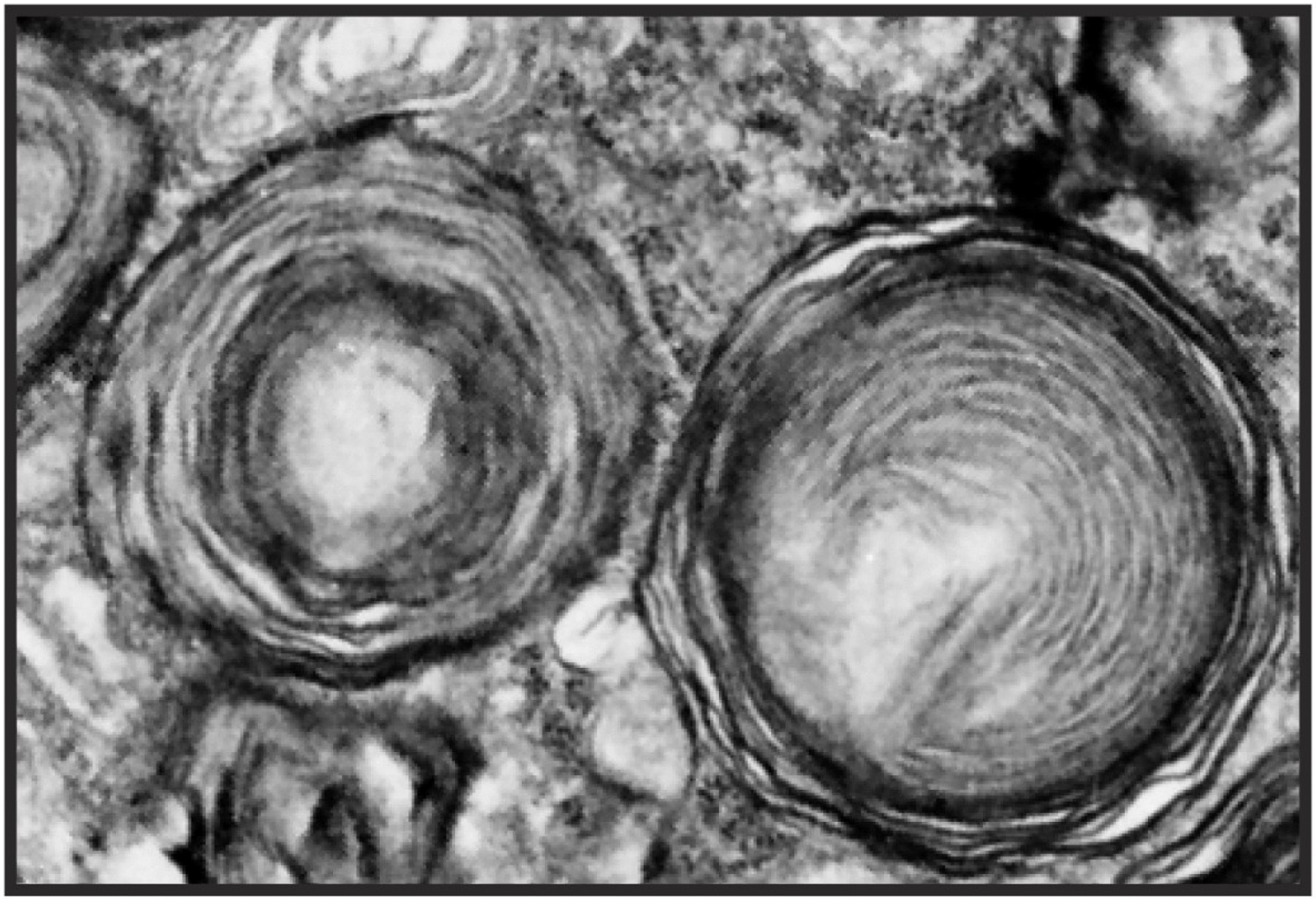
Figure 10-15
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

Defects in the turnover of membrane lipids lead to a number of diseases



Box 10-1 figure 1
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

Tay-Sachs lysosomes



Sterols and Cholesterol

- Sterol
  - Steroid nucleus: four fused rings
  - Hydroxyl group (polar head) in the A-ring
  - Various nonpolar side chains
- The steroid nucleus is almost planar

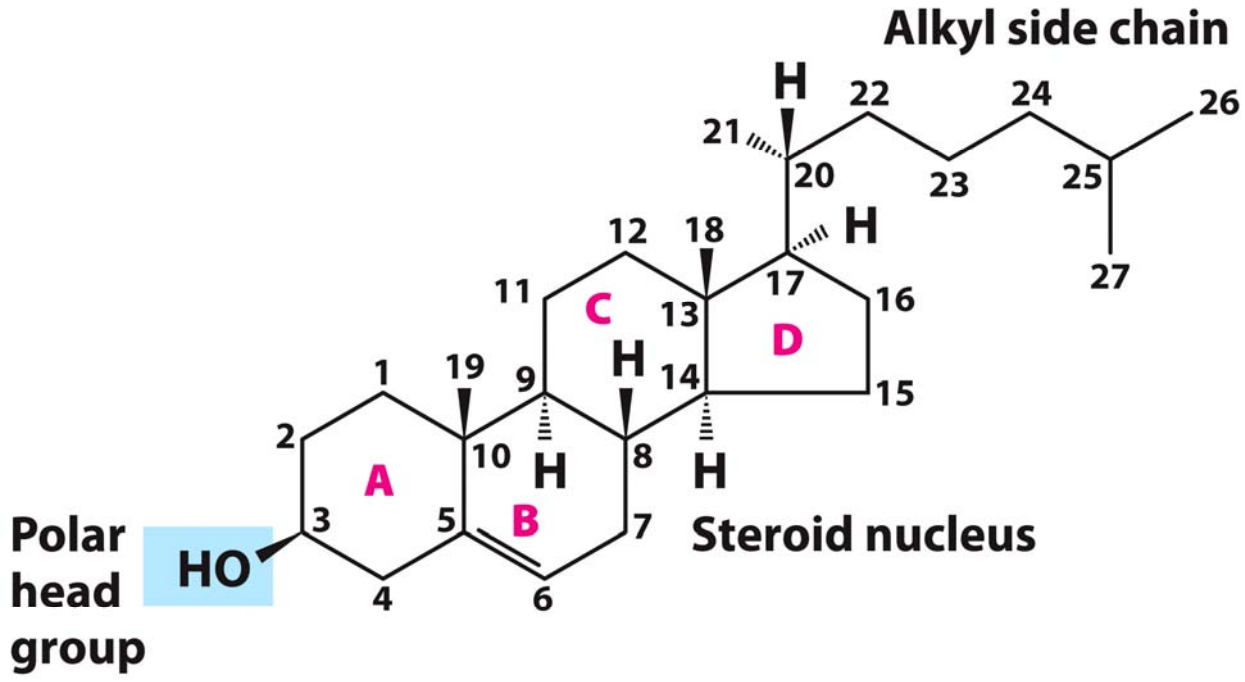


Figure 10-17
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

Physiological Role of Sterols

- Cholesterol and related sterols are present in the membranes of most eukaryotic cells
  - Modulate fluidity and permeability
  - Thicken the plasma membrane
  - Most bacteria lack sterols
- Mammals obtain cholesterol from food or synthesize it *de novo* in the liver
- Cholesterol, bound to proteins, is transported to tissues via blood vessels
  - Cholesterol in low-density lipoproteins tends to deposit and clog arteries
- Many hormones are derivatives of sterols

Membrane Composition Is Highly Variable in Different Organelles

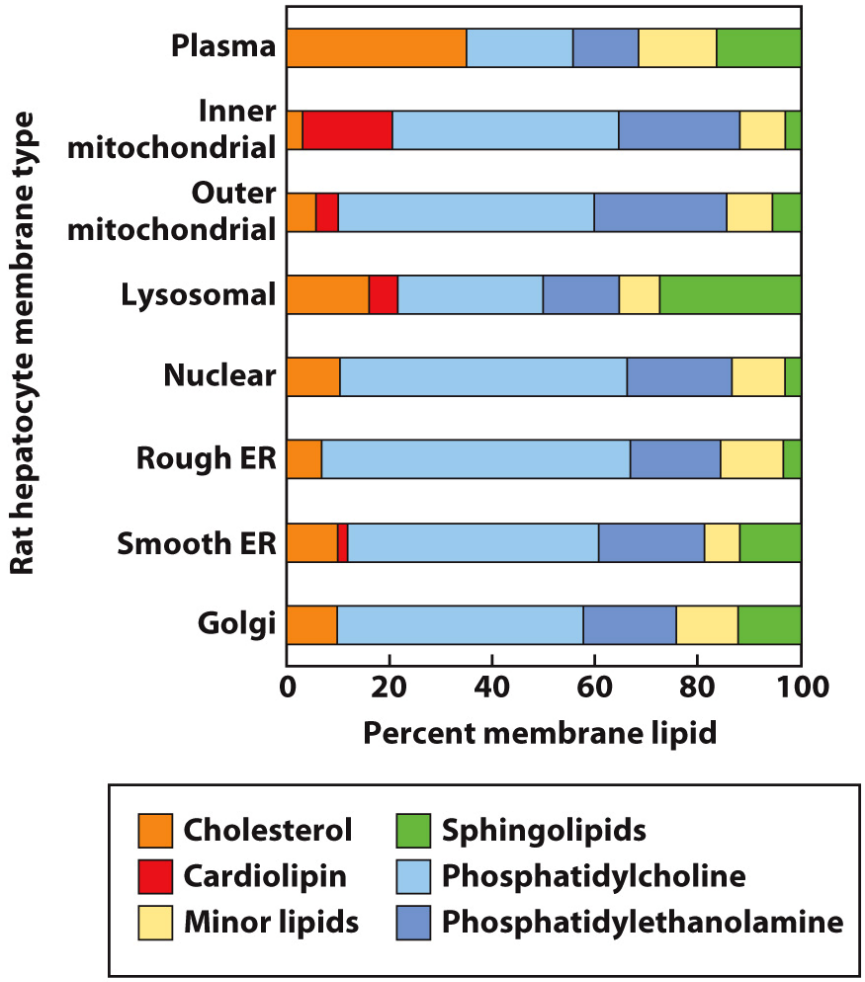


Figure 11-2
Lehninger Principles of Biochemistry, Seventh Edition
© 2017 W. H. Freeman and Company

Membrane Rafts

Lipid distribution in a single leaflet is not random or uniform.

- Lipid rafts:
  - contain clusters of glycosphingolipids with longer-than-usual tails
  - are more ordered
  - contain specific doubly or triply acylated proteins
  - allow segregation of proteins in the membrane

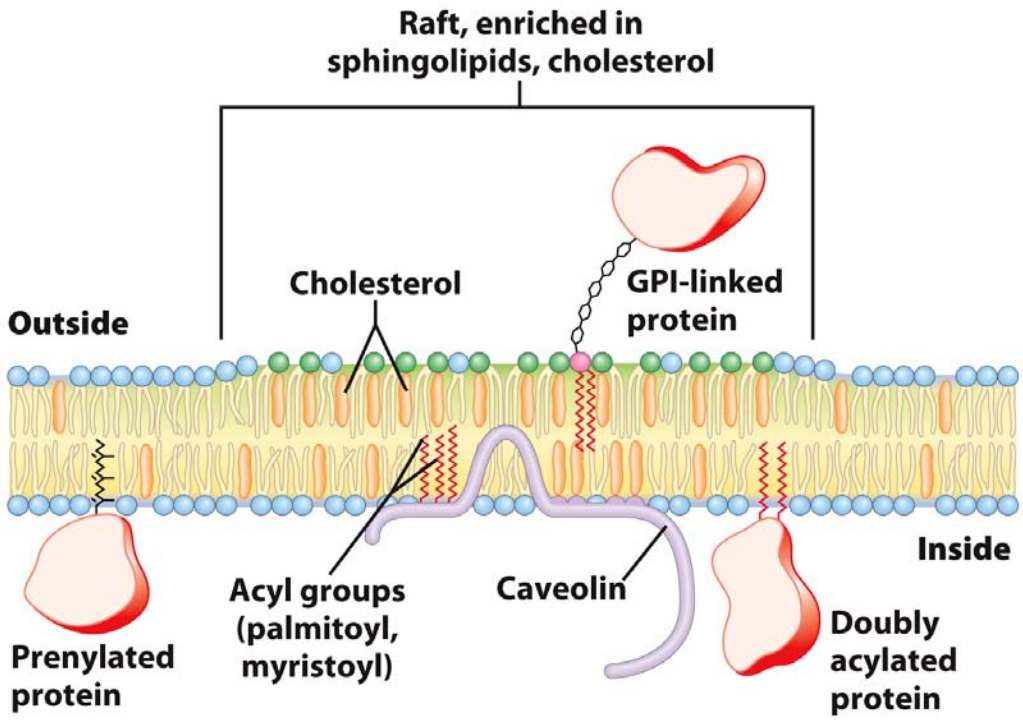
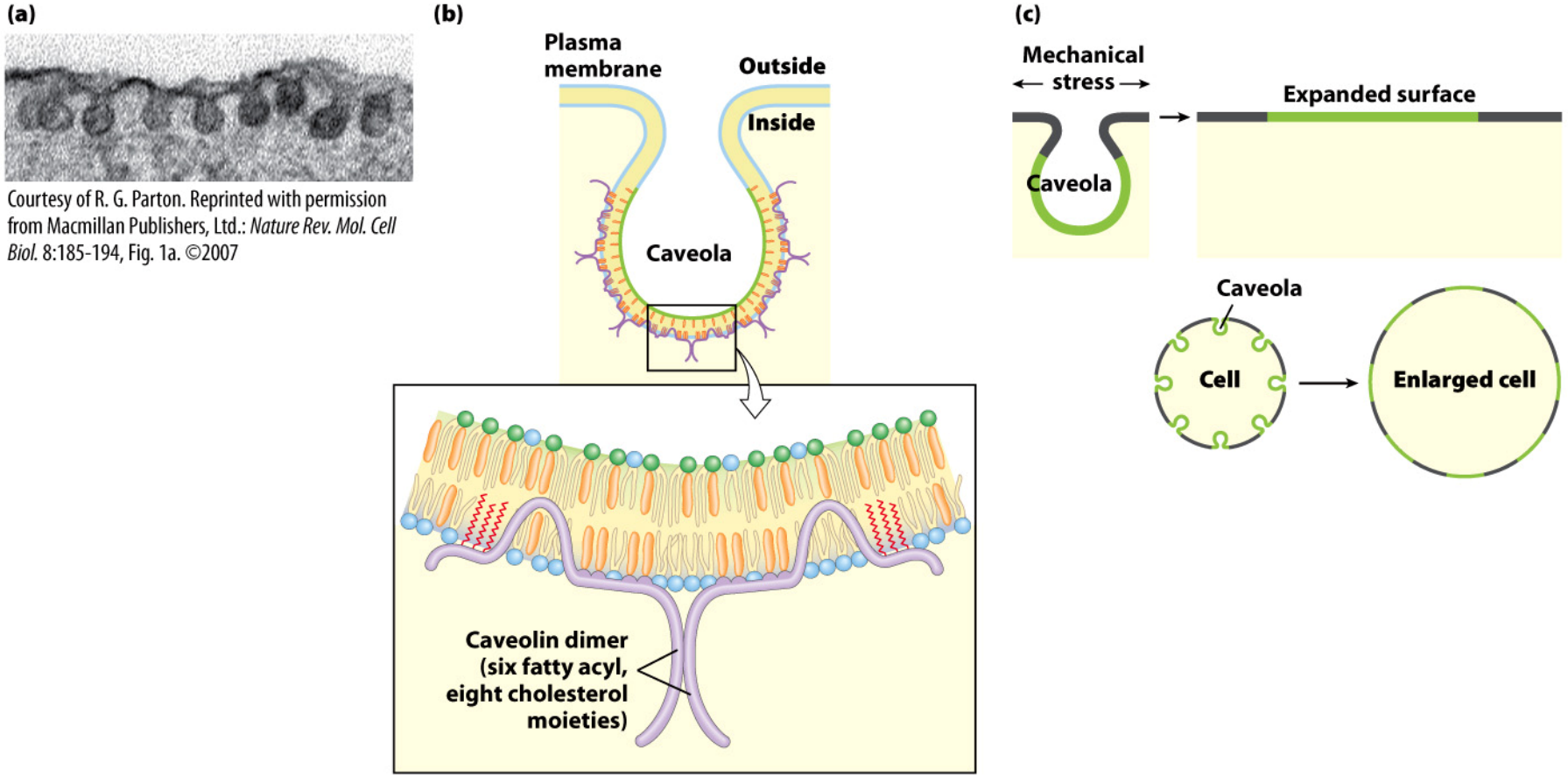


Figure 11-19
Lehninger Principles of Biochemistry, Seventh Edition
© 2017 W. H. Freeman and Company

Caveolin Forces Membrane Curvature



Courtesy of R. G. Parton. Reprinted with permission from Macmillan Publishers, Ltd.: *Nature Rev. Mol. Cell Biol.* 8:185-194, Fig. 1a. ©2007

Figure 11-20
Lehninger Principles of Biochemistry, Seventh Edition
© 2017 W. H. Freeman and Company